Modern Distribution of Condemned Buildings in Pittsburgh with Historical Redlining Maps

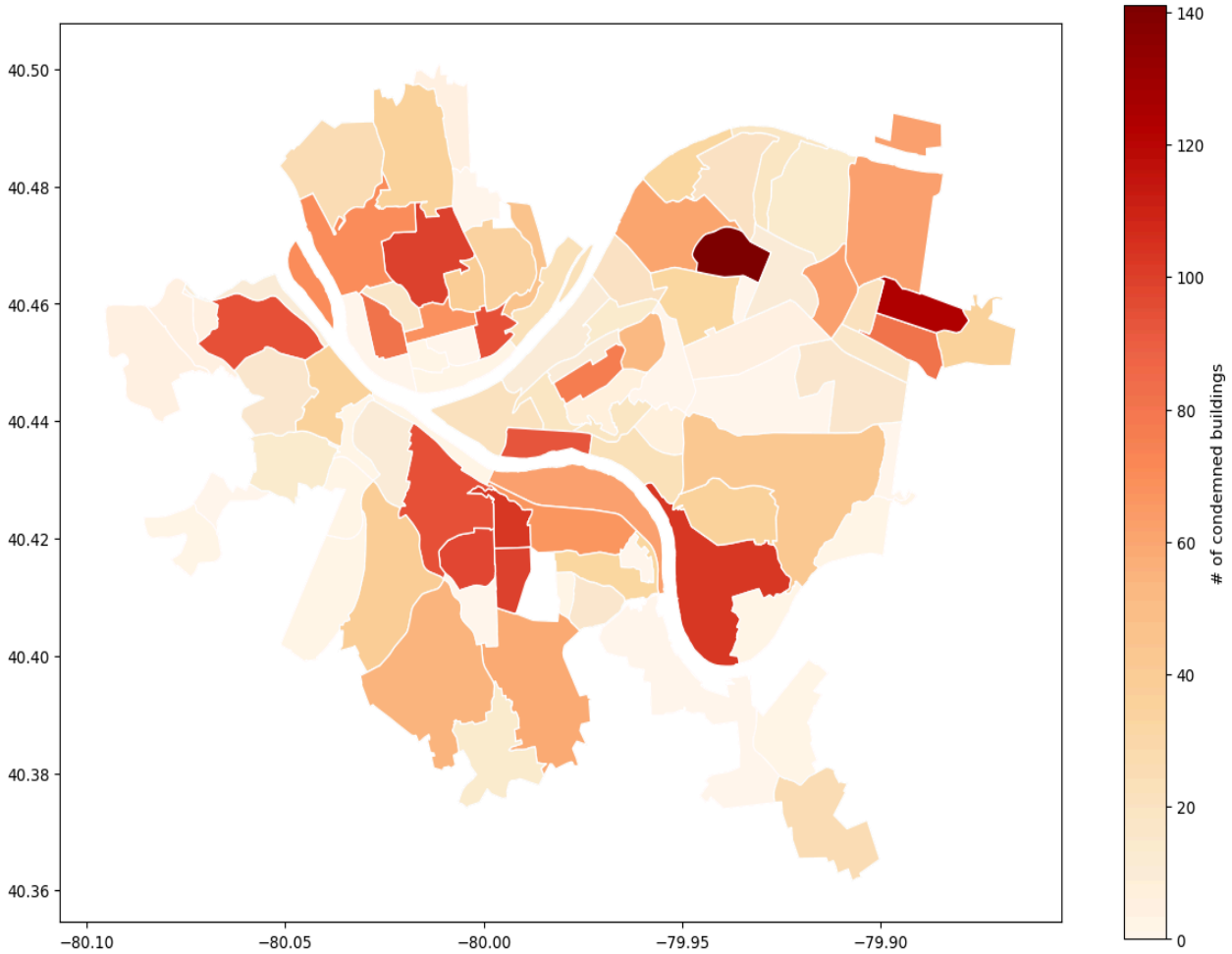
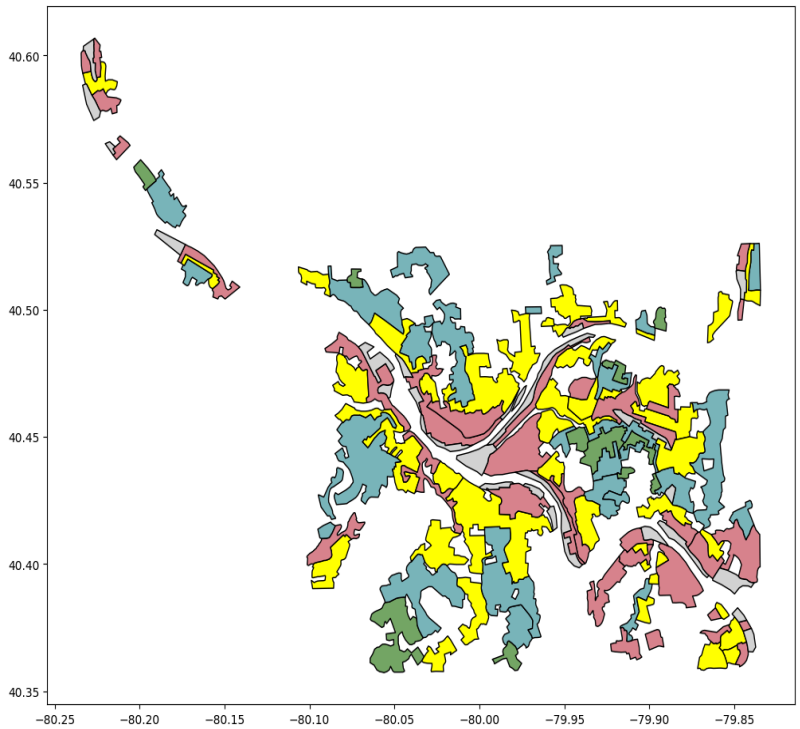


Fig. 1 (left) This heatmap shows the number of condemned buildings per neighborhood. The legend explains that the maximum # of condemned buildings in any neighborhood is 140, and the darker the color, the more condemned buildings in the neighborhood.

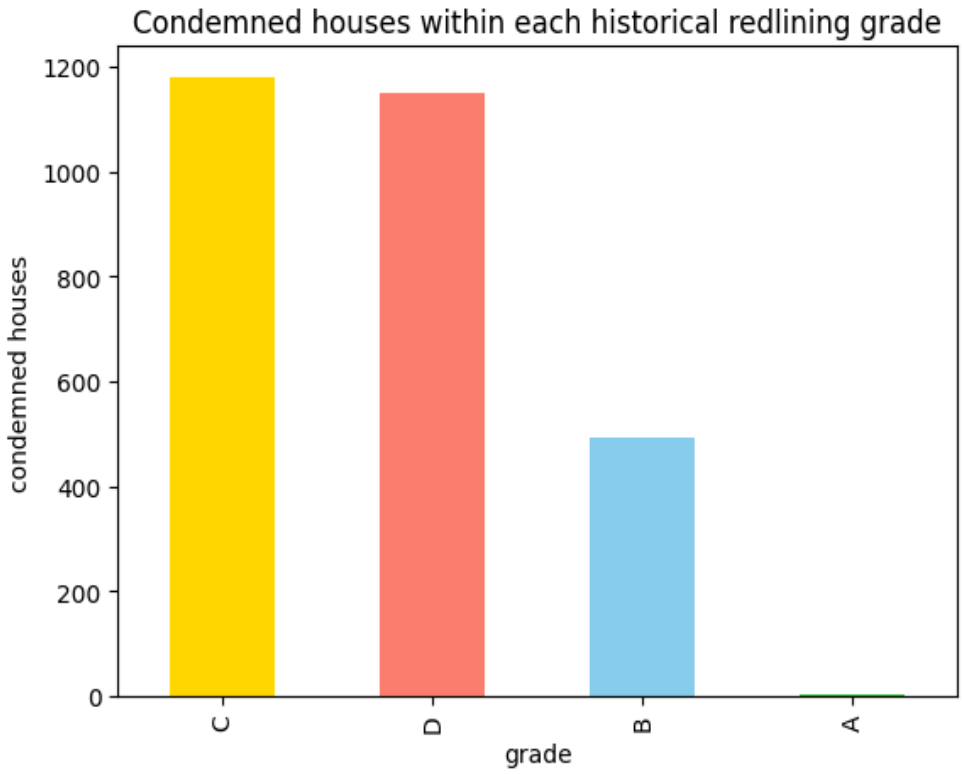
Fig. 2 (right) This is a version of a map created by the HOLC (Home Owners Loan Corporation) in 1937, color-coding neighborhoods for the purposes of controlling housing loans. Red meant mostly nonwhite residents, yellow and blue were racially mixed, and green meant mostly white residents. Green neighborhoods were most likely to receive mortgage loans, then blue, then yellow, and red neighborhoods essentially never. [1]



[1] Nelson, R. K., Winling, L, et al. (2023). Mapping Inequality: Redlining in New Deal America. Digital Scholarship Lab. https://dsl.richmond.edu/panorama/redlining.

Findings

Combining these two datasets confirms that the majority of modern condemned buildings in Pittsburgh neighborhoods are in historically redlined neighborhoods.



Percentages:

- 40% in D
- 41% in C
- 18% in B
- 1% in A

Significance

Due to this documented history of systemic marginalization, racial minorities in Pittsburgh (primarily Black Pittsburghers) continue to face difficulty obtaining mortgage loans to build generational wealth [2], which negatively affects health on a long-term scale[3]. This is due in part to a withholding of resources from historically Black neighborhoods, which is reflected in Fig. 1 as modern crumbling architectural infrastructure, while historically wealthier areas receive necessary grants and funding or don't require them due to consistent upkeep and high property taxes. The health and prosperity of a city's population are both incredibly important to the daily lives of citizens in a city and to the city's allocation of funds and resources. Understanding where to aim money is integral to a functioning democracy, which is why reports like the Housing Needs Assessment have similar figures corroborating the takeaway that infrastructure is worse in majority-Black neighborhoods, which also tend to be primarily redlined neighborhoods that have been subjected to urban development and renewal projects. Pittsburgh is a notably segregated city and it was reported 7 years ago as having worse quality-of-life for Black residents than other similar cities[4]. The government-level data analysis being directed towards this subject is reflective of this need for change and improvement on a larger scale.

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[2] City of Pittsburgh. (2022) *Pittsburgh Housing Needs Assessment*. Department of City Planning. https://hdp-us-prod-app-pgh-engage-files.s3.us-west-2.amazonaws.com/3516/9765/4629/Pittsburgh_HNA_Final_Report_01312022_1.pdf

[3] Jones, Emily J et al. (2024). Historical Structural Racism in the Built Environment and Physical Health among Residents of Allegheny County, Pennsylvania. *Journal of urban health : bulletin of the New York Academy of Medicine*, *101*(4), 713–729. https://doi.org/10.1007/s11524-024-00884-6

[4] Howell, Junia & Goodkind, Sara & Jacobs, Leah & Branson, Dominique & Miller, Liz. (2019). Pittsburgh's Inequality Across Gender and Race. 10.13140/RG.2.2.24443.46883.

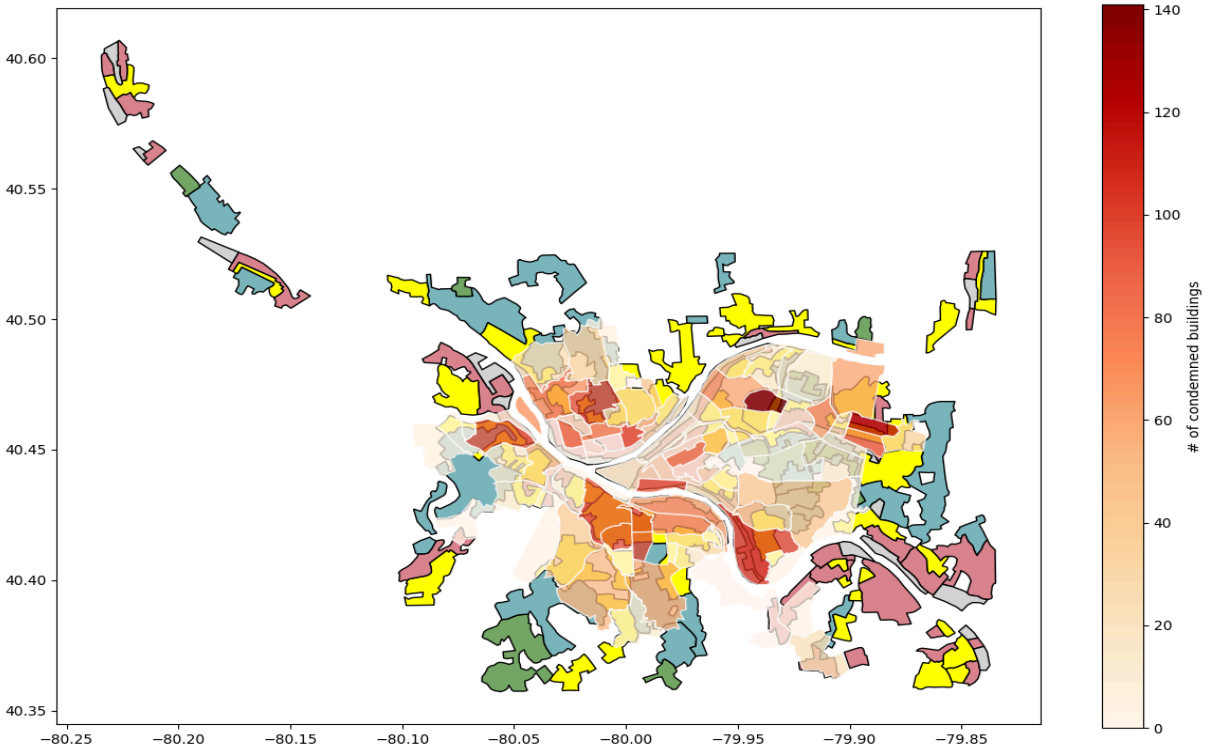

Fig. 3 (left) Figures 1 and 2 overlaid, showing a visual overlap in addition to the shown statistical overlap between the historically redlined neighborhoods and modern building stability information.

Data and Methodology

This visualization was created with open-source shape files and neighborhood data from the City of Pittsburgh and filled in with the condemned building dataset, both sourced from the Western Pennsylvania Regional Data Center, which classmate Javin Faisant and I found together and used for different purposes, as I confirmed with. The parcels in the condemned building dataset were mapped onto corresponding latitudes and longitudes with data from WPRDC's Parcels N'at bulk downloader program, from a Permits Licenses and Inspections (PLI) dataset. The latitudes and longitudes attached to each parcel allowed for the combination of the condemned buildings data with both the Pittsburgh neighborhood data and the redlining data from the Mapping Inequality interactive database compiled by University of Richmond. With additional data from Pittsburgh census tracts [5] to fill in neighborhood data, the creation of a GeoPandas heatmap-style plot was possible. Mapping the redlining data utilized the same concept, as it was open-source and easy to acquire the JSON files, and they had the same crc.

Repo: https://github.com/kilometrees/final-project-information-visualization.git

[5] Lemery, Christopher. *Pittsburgh Census Information @ Pitt: Pittsburgh Census Tracts, 1940-2020.* University of Pittsburgh Library System. (Updated 2025). https://pitt.libguides.com/pghcensus/pghcensustracts